

1.1 Problem Statement

Agriculture forms the foundation of many economies around the world, providing essential food supplies and employment. However, crops are frequently threatened by various diseases that reduce productivity and quality. Specifically, banana plants, a vital tropical crop, are highly susceptible to diseases such as Panama Disease, Black Sigatoka, and Banana Bunchy Top Virus.

Early and accurate detection of these diseases is critical to minimize yield loss and to ensure food security. Traditional methods of disease detection involve manual inspection, which is often time-consuming, error-prone, and inaccessible to small farmers. The lack of expertise in rural areas further compounds the problem. Thus, there is a pressing need for an intelligent system that can automatically detect and predict crop diseases accurately and in real time.

The proposed project, AI-Driven Crop Disease Prediction and Management System, specifically focuses on banana plants and aims to leverage artificial intelligence techniques to identify diseases early based on image analysis. The goal is to help farmers take timely and appropriate actions to manage the disease, thus improving agricultural productivity and sustainability.

1.2 Objectives

The main objectives of the proposed system are:

- To develop a machine learning-based model capable of predicting banana plant diseases from leaf images with high accuracy.
- To create a user-friendly system that can be used by farmers with minimal technical knowledge.
- To explore and implement deep learning models like CNN, MobileNetV2, EfficientNet, and DenseNet for disease detection.
- To reduce crop loss by facilitating early detection and management of plant diseases.
- To promote environmentally sustainable farming by reducing unnecessary pesticide usage through targeted disease management.

1.3 Applications

The AI-Driven Crop Disease Prediction System has various practical applications:

- Agricultural Assistance: Helps farmers quickly identify diseases and take corrective measures.
- Agri-Tech Startups: Can be integrated into mobile applications for agricultural technology companies.
- Government and NGOs: Useful for rural development programs aimed at enhancing farming practices.
- Research and Development: Provides a large dataset for future studies in plant pathology and AI in agriculture.
- Educational Tool: Can be used in agricultural universities to teach students about plant diseases and the use of AI in modern farming.